IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-130

1. Topic (12 words max)

Interactive Learning tool for Dyslexia Kids

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-130
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Dyslexia is a neurodevelopmental learning difficulty affecting 5-10% of children globally, characterized by persistent challenges in accurate and fluent word reading, spelling, and phonological processing despite adequate instruction [1].

In early primary years (ages 5-8), dyslexic learners exhibit slow decoding, frequent reading errors, inconsistent spelling and reduced classroom confidence [1]. Research identifies four interconnected skill deficits underlying these difficulties: phonological awareness (perceiving and manipulating speech sounds), spelling/encoding (mapping sounds to written letters), reading/decoding (accurate word recognition and fluency), and working memory & sequencing (holding and ordering information during cognitive tasks) [1][3].

Approximately 60-80% of dyslexic children experience working memory weaknesses that limit their ability to blend phonemes, remember letter sequences, follow multi step instructions, and retain meaning while reading, creating a "cognitive bottleneck" that slows literacy acquisition even with appropriate teaching [3][4]. Evidence based intervention requires structured literacy instruction explicit, systematic, cumulative teaching with frequent corrective feedback and mastery based practice [2] yet access to individualized, comprehensive support remains limited, particularly in resource constrained settings like Sri Lanka, where teacher awareness and intervention tools are inadequate [5][8].

Digital game-based learning offers promise by delivering engaging, adaptive practice with immediate feedback; studies demonstrate that game-based phonics interventions improve print knowledge and decoding in at risk young learners [6][7].

However, existing dyslexia apps focus predominantly on phonics and reading support, rarely integrating explicit working memory training as an equal component, and are typically Western-designed without local curriculum alignment or bilingual (Sinhala-English) capabilities [5][8]. Therefore, the research problem is: How can an interactive, web-based gamified learning platform be designed and evaluated to address all four interconnected skill domains phonological awareness, spelling/encoding, reading/decoding, and working memory & sequencing through structured, mastery based games that support the complete cognitive and literacy profile of dyslexic children aged 5-8 in the Sri Lankan educational context

References

[1] International Dyslexia Association. (2020). "Effective Reading Instruction for Students with Dyslexia." Available at: https://dyslexiaida.org/effective-reading-instruction-for-students-with-dyslexia/

[2] International Dyslexia Association. (2017). "What Is Structured Literacy?" Available at: https://dyslexiaida.org/what-is-structured-literacy/

[3] International Dyslexia Association. (2020). "Working Memory: The Engine for Learning." Available at: https://dyslexiaida.org/working-memory-the-engine-for-learning/

[4] British Dyslexia Association. (2025). "How Does Dyslexia Affect Maths Learning?" Available at: https://www.bdadyslexia.org.uk/dyscalculia/how-can-i-identify-dyscalculia/how-does-dyslexia-affect-maths-learning

[5] Repository, Sabaragamuwa University. "LEXIS: A Preliminary Screening and Intervention Tool for Children with Dyslexia." Available at: http://repo.lib.sab.ac.lk:8080/xmlui/handle/123456789/1757

[6] Vasalou, A., et al. (2021). "Digital Game-Based Phonics Instruction Promotes Print Knowledge in Pre-Readers at Cognitive Risk for Dyslexia." Frontiers in Psychology, 12, 720548. https://doi.org/10.3389/fpsyg.2021.720548

[7] PubMed. (2021). "Digital Game-Based Phonics Instruction Promotes Print Knowledge in Pre-Readers at Cognitive Risk for Dyslexia." PMID: 34566803. Available at: https://pubmed.ncbi.nlm.nih.gov/34566803/

[8] Sri Lanka Clinical Trials Registry. (2023). "Effectiveness of a Multisensory Interactive Sinhala Alphabet Learning Tool." Available at: https://slctr.lk/trials/slctr-2016-016

6. Brief description of Existing Research and Systems (300 words max)

International Products and Systems

- Platforms like **Lexercise** and **Dyslexia Games** offer short, daily game-based sessions focused on phonological training and reading fluency through structured exercises and feedback.
- **GraphoGame** targets phoneme-grapheme mapping using adaptive algorithms that adjust difficulty based on learner performance.
- **Nessy** uses multisensory literacy instruction, combining visual, auditory, and kinesthetic elements to support phonics and decoding.
- These tools demonstrate that structured, digital, game-based practice supports phonological awareness effectively but are designed for English-speaking contexts without cultural adaptation for Sri Lankan learners.

Sri Lankan Products and Systems

- **ARUNALU** (2020, SLIIT) is a mobile learning ecosystem targeting Sinhala reading weaknesses in children aged 5-6, using voice recognition and natural language processing for phonological training and automated screening.
- **LEXIS** (Sabaragamuwa University, 2021) provides web-based preliminary screening and basic intervention modules for dyslexia, focusing on reading and spelling with interactive learning activities.
- **Multisensory Interactive Sinhala Alphabet Tool** (Clinical Trial, Gampaha District) uses visual and phonological reinforcement to improve letter recognition and reading fluency in children with dyslexia.
- **Dyslexia Learning Assistant for Sinhala Children** (2024, mobile app) focuses on Sinhala handwriting practice with text-to-speech, machine learning-based handwriting recognition, and real-time feedback for writing skills.
- **Low-Resource Speech-Driven NLP Pipeline** (2024) is a real-time speech-driven assistive system for Sinhala-speaking individuals with dyslexia, integrating speech-to-text, error correction, and comprehension support.

Gap of Existing Research

- **Bilingual support**: Existing tools (Lexercise, GraphoGame, Nessy) are English-only; local tools (ARUNALU, LEXIS) lack comprehensive integration
- **Working memory training**: Most dyslexia platforms focus only on reading/phonics and ignore working memory a core cognitive deficit
- **Four-domain integration**: No single platform addresses phonological awareness, spelling, reading, AND working memory together
- **Cultural adaptation**: Limited tools use Sri Lankan curriculum content and cultural context
- **Advanced analytics**: Absence of combined eye-tracking, brain processing speed assessment, and pronunciation analysis in one system

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an interactive, web-based gamified learning platform designed to provide comprehensive, adaptive intervention for children aged 5-8 with dyslexia in the Sri Lankan context. It integrates four interconnected skill domains-phonological awareness, spelling/encoding, reading/decoding, and working memory & sequencing within a structured literacy framework, delivering mastery-based, personalized learning through engaging game mechanics.

1. **Initial Assessment Module**

- Conducts baseline screening across all four skill domains using adaptive testing that adjusts difficulty in real time.
- Generates an individualized learner profile highlighting strengths, weaknesses, and priority areas for intervention.
- Supports bilingual (Sinhala-English) assessment aligned with local educational requirements.

2. **Adaptive Game-Based Intervention Engine**

- Delivers structured, cumulative learning activities through age-appropriate, culturally adapted games mapped to the Sri Lankan primary curriculum.
- Dynamically adjusts difficulty, content sequencing, and skill focus based on ongoing performance data.
- Provides immediate corrective feedback, audiovisual cues, and multisensory reinforcement, with mastery-based progression so learners must demonstrate competence before advancing.

3. Skill-Specific Game Modules *(Four member's components)*

- **Phonological Awareness:** Targets the ability to connect letters with their corresponding sounds (letter–sound mapping), construct words from individual letters or sounds, and practice recognizing and applying common spelling patterns.
- **Pronunciation & Speech Sound Learning:** Develops accurate phoneme production and articulation through listen–repeat–compare mechanics using native Sinhala speech samples. Analyzes speech at the individual sound level, provides multimodal feedback (visual cues and audio playback) to correct pronunciation errors, and tracks specific phoneme difficulties for personalized practice pathways
- **Reading/Decoding:** Develops word recognition abilities, builds reading fluency through timed exercises, and provides scaffolded comprehension activities to support understanding of written text.
- **Working Memory & Sequencing:** Strengthens core executive functions through digit and item span tasks (remembering sequences of numbers or objects), pattern recall exercises, multi-step instruction-following activities, and sequencing tasks that train the brain to hold and manipulate information

4. **Progress Tracking and Analytics Dashboard**

- Visualizes individual learning trajectories, error patterns, and mastery levels for teachers and parents.
- Produces automated progress reports with practical recommendations for instructional adjustments.
- Monitors engagement, time on task, and response accuracy to refine each learner's pathway.

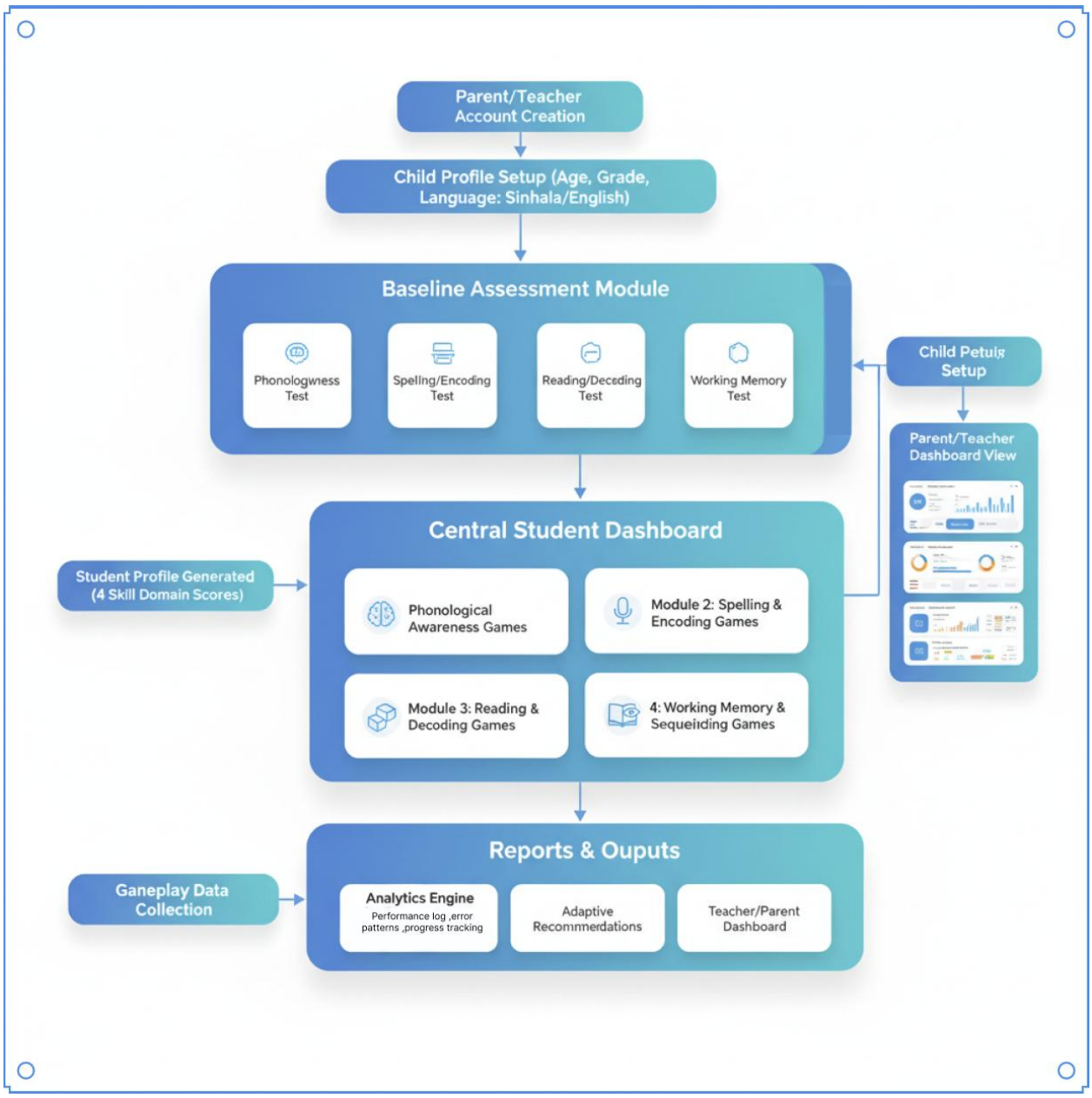
5. **Gamification System**

- Uses points, badges, levels, and unlockable rewards to maintain motivation and persistence.
- Supports adaptive gamification by tuning reward frequency and challenge based on learner behavior and preferences

The platform is a secure, web-based system that works on multiple devices. It uses structured, multisensory teaching with step-by-step lessons, regular feedback, and appropriate difficulty levels to help learners build confidence and skills effectively.

In short input process output flow.

- **Inputs:** Child profile (age, grade, language), baseline assessment scores, and ongoing gameplay data (responses, time, errors).
- **Processes:** Adaptive assessment, game selection and difficulty adjustment, skill training in the four domains, analytics, and gamification.
- **Outputs:** Learner profile and skill scores, real-time dashboards, progress reports, and updated personalized learning pathways.


Parent/Teacher Account Creation
Child Profile Setup (Age, Grade, Language: Sinhala/English)
Baseline Assessment Module
Phonologwness Test
Spelling/Encoding Test
Reading/Deceding Test
Working Memory Test
Child Petuig Setup
Parent/Teacher Dashboard View
Central Student Dashboard
Student Profile Generated (4 Skill Domain Scores)
Phonological Awareness Games
Module 2: Spelling & Encoding Games
Module 3: Reading & Decoding Games
4: Working Memory & Sequeiding Games
Reports & Ouputs
Ganeplay Data Collection
Analytics Engine
Performance log ,error patterns ,progress tracking
Adaptive Recommerdations
Teacher/Parent Dashboard

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain Expertise in Dyslexia and Structured Literacy

The project team will acquire specialized domain knowledge through a systematic review of academic literature related to dyslexia and structured literacy. This includes studies on phonological awareness, reading and spelling difficulties, pronunciation challenges, and working memory deficits commonly seen in dyslexic children. By reviewing this literature, the team will gain a strong understanding of evidence based intervention methods, structured literacy principles, and multisensory learning techniques that are widely used to support children with dyslexia. This theoretical foundation will guide the design of learning activities and ensure that the platform follows scientifically proven approaches.

Local domain expertise will be further strengthened through direct consultations with special education teachers, speech-language therapists, and educational psychologists who have experience working with Sri Lankan children with learning difficulties. These professionals will provide practical insights into how dyslexia presents in real classroom environments and how children respond to different teaching strategies. Collaboration with the Dyslexia Association of Sri Lanka and researchers involved in local projects such as LEXIS and Arunalu will help ensure that the platform is culturally appropriate and aligned with the Sri Lankan education system. These interactions will also support the identification of language-specific challenges faced by Sinhala- and English-medium learners and ensure alignment with local curriculum requirements and classroom practices.

Technical Knowledge and Skills Development

Frontend development skills using React.js or Vue.js and backend development skills using Node.js or Django and CoAI. This technical knowledge will support the development of a secure, responsive, and user-friendly web-based platform that can be accessed across different devices.

Gamification knowledge will be developed using browser-based game development frameworks such as Unity WebGL or Phaser.js. The platform will incorporate adaptive difficulty levels, reward mechanisms, and interactive feedback to maintain engagement and motivation among children with learning difficulties. Speech recognition features for both Sinhala and English will be implemented using web-based speech technologies to support listening and pronunciation activities. In addition, data analysis and visualization skills using Python libraries and dashboard tools will enable effective monitoring of learner performance, progress tracking, and system evaluation.

Data Collection and Ethical Considerations

Primary data will be collected directly from selected primary schools with the cooperation of teachers and school administrators. Age appropriate screening activities embedded within the platform will be used to collect baseline performance data from the general student population. This approach allows the identification of children who may show dyslexic characteristics based on learning patterns without requiring a prior clinical diagnosis.

Collected data will include accuracy rates, response times, error patterns, and task completion behavior across reading, spelling, phonological awareness, and working memory activities. Intervention data will track engagement levels, skill mastery, level progression, and learning improvement over time. Additional demographic information such as age, grade level, and language preference will be collected, along with qualitative feedback from teachers and parents to support interpretation of results.

All data collection activities will be conducted only after obtaining ethical approval, parental consent, child assent, and official school permissions. Strong data protection measures, including anonymization, secure data storage, and controlled access, will be implemented to ensure participant privacy and ethical compliance throughout the research process.

9. Objectives and Novelty

Main Objective To design, develop, and validate a web-based gamified learning platform for dyslexic children aged 5-8 in Sri Lanka that addresses all four core skill areas phonological awareness, spelling, reading, and working memory-through bilingual (Sinhala-English) games with adaptive difficulty and personalized learning pathways.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Dewminie W.T.M *IT 22127150*	Memory & Sequencing Module (Working Memory Support)	• Develop memory-based games using Sri Lankan Grade 1-3 curriculum content such as animals, fruits, days of the week, months, and the alphabet • Create digit-span activities and "What's Missing?" games where the level of difficulty gradually increases based on the child's performance • Design multi-step instruction-following games (e.g., "tap, clap, jump") to help children practice remembering and completing sequences of actions • Implement an adaptive learning system that automatically adjusts the memory load (e.g., from 2 to 3 to 4 items) according	• **Focus on underlying cognitive skills**: The platform directly targets working memory. • **Curriculum-aligned learning**: All activities are based on Sri Lankan Grade 1-3 syllabus content, allowing children to strengthen school related knowledge while improving memory skills • **Instruction-following skill development**: The system trains children to remember and follow multi-step instructions, addressing a common classroom challenge not usually covered in dyslexia applications

		to the child's accuracy and consistency • Implement scaffolded error correction: when children fail, show correct answers with visual/audio reinforcement, then provide similar questions with decreasing support until mastery • Develop a progress-tracking mechanism to record accuracy, error patterns, attempts, and improvement across all activities	• **Error pattern remediation**: System identifies specific failure points, reveals correct answers, then generates targeted similar practice until consistent accuracy ensuring dyslexic learners consolidate skills through guided repetition • **Mastery-based progression**: Learners advance only after demonstrating consistent success, ensuring that skills are properly reinforced before moving to higher levels
Fernando K.H.D.M IT22141088	Phonological Awareness & Spelling Module	• Design drag-and-drop "sound block" manipulation interface for letter tiles and word construction. • Create "change one sound" puzzles with integrated audio support and visual feedback. • Build spelling construction activities with minimal text dependency and audio cues.	• **Audio-heavy, text-light design:** Unlike traditional reading-dependent spelling apps, this module uses heavy audio so children with severe reading difficulties can still practice phonological skills without needing to read instructions. • **Sinhala-English sound mapping:** Includes games that explicitly connect similar

		• Develop error pattern detection system for consonant confusion (b/d, p/q reversals).	phonemes across languages (e.g., "ಪ" to "ba") - not found in existing English-only dyslexia tools.
Khan N.A.B IT22266064	Eye-Tracking–Based Reading Behaviour and Brain Processing Speed Assessment for Dyslexic Children	• Design **eye-tracking-enabled reading tasks** to capture gaze behaviour while children interact with letters, words, and short visual reading stimuli. • Develop **simple, level-based cognitive games** to measure brain processing speed, visual reaction time, and attention switching. • Integrate **web-camera–based eye tracking** to record fixation duration, gaze shifts, regressions, and attention loss during reading and game activities. • Implement algorithms to calculate **brain speed indicators**	• **Combined Eye-Tracking and Brain-Speed Assessment:** Unlike existing dyslexia tools that focus only on reading accuracy, this component integrates eye-tracking during reading with cognitive game performance to assess how quickly and efficiently children process visual information. • **Brain Processing Speed through Gameplay:** Introduces level-based, child-friendly games to estimate brain processing speed using reaction time and gaze-based decision metrics instead of

		such as reaction time, response delay, and gaze-based decision time across different game levels. • Design **adaptive game levels** that gradually increase visual and cognitive complexity to observe changes in processing speed. • Log and analyze eye-movement and game-interaction data in real time for behavioural analysis. • Classify children's **visual processing and brain speed levels** (Low / Medium / High) based on combined eye-tracking and game performance data. • Generate structured outputs to support **personalized learning and early intervention decisions** (non-diagnostic).	traditional test-based measurements. • **Low-Cost Web-Based Eye Tracking:** Uses standard web cameras for eye-tracking, avoiding expensive medical hardware and making the solution practical for Sri Lankan schools. • **Behaviour-Oriented Analysis:** Focuses on visual attention patterns, hesitation, and processing delay, rather than only correct or incorrect answers. • **Adaptive Learning Support:** Brain speed and eye-tracking insights dynamically inform task difficulty and visual complexity, enabling personalized learning pathways.

Navod J.N IT22027542	Pronunciation & Speech Sound Learning Module.	• Develop listen–repeat–compare mechanics using slow, clear native Sinhala speech samples • Implement phoneme-level speech input capture via mobile/web microphone access • Build pronunciation practice flows with syllable segmentation and gradual sound combination • Log pronunciation attempts and error patterns for adaptive lesson sequencing • Integrate visual and audio feedback mechanisms suitable for dyslexic early learners.	• **Phoneme-Level Pronunciation Error Diagnosis**: Unlike existing dyslexia learning tools that evaluate pronunciation at the word level, this module analyzes children's speech at the individual phoneme level, enabling precise identification of specific mispronounced sounds rather than marking entire words as incorrect. • **Error-Pattern–Driven Adaptive Remediation:** The system records pronunciation error patterns over time and dynamically adapts future practice activities to focus on the child's most persistent speech sound difficulties, creating a personalized pronunciation learning pathway. • **Closed Feedback Loop Between Assessment and Intervention:** Pronunciation assessment results are directly linked to lesson

			sequencing, ensuring that detected speech sound weaknesses automatically influence subsequent learning tasks instead of following a fixed lesson order. • **Foundational Speech-to-Literacy Support:** By targeting speech sound production accuracy at the phoneme level, the module strengthens foundational oral language skills that directly support later reading and spelling development, aligning with structured literacy principles recommended by the International Dyslexia Association.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

The reserach topic & functions are accepted

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Sanvitha	Kasthuriarachchi	
Co-Supervisor	Miss	Thisara	Shyamalee	
External Supervisor	Dr.	Kamalani	Wanigasinghe	Dr. W. M. K. [illegible] NIGASINGH (M.B.B.S.) S.L.M.C. R[illegible]. No: 11614 MO[illegible] ABE
Summary of external supervisor's (if any) experience and expertise				

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